PROJECT DELIVERY BRIEF / FLYRANK AI BACKEND

El-Mizan Background Job API

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What I built

I moved El-Mizan's slow national-ID request classification out of the web request and into a background job. The API accepts a request quickly, returns a report ID with **202 Accepted**, and lets the caller check progress until the worker finishes. The worker connects to the existing A6 classifier when a model key is configured.

REQUEST	BACKGROUND WORK	RESULT
POST /reports Returns 202 + report ID	Inngest worker waits durably for 8 seconds, then runs classification or an explicitly labelled no-key demo.	GET /reports/{id} pending → running → done / failed

Reliability and safety

- **Duplicate protection:** stable event IDs and an Idempotency-Key prevent a repeated submit from creating a second report; reusing a key with different content returns 409.
- **Failure handling:** the worker retries twice (up to three attempts). Terminal failures update report status and emit a privacy-safe alert; an optional webhook can be configured.
- **Operational controls:** concurrency is capped at two jobs and a heartbeat logs pending, running, completed, and failed counts every minute.
- **Input controls:** request body and topic length are bounded; malformed or missing topics are rejected before enqueueing.

Verification

10 automated tests passed. I also ran the actual local Inngest Dev Server end to end: a submitted report returned 202 pending and reached done after the durable wait. A forced failure exhausted retries and reached failed, with the alert callback invoked. The successful test had no model API key configured; its result explicitly states AI was skipped, so this verifies queue-to-status flow rather than a live model response.

Availability and next step

The portfolio website is publicly reachable. The El-Mizan API and Inngest worker are a verified local prototype and are **not deployed as a public API**. Report state is currently held in memory, so production deployment would require persistent storage and configured provider credentials.

Live portfolio
hany-s-world-597716960036.europe-west2.run.app
Checked: HTTP 200 on 2026-10-13

Assignment source
GitHub: background-job folder
Six staged commits plus verified-run update

Submission note: code, tests, build log, and real run IDs are in the GitHub folder. The required raw, unedited desktop screen recording still needs to be captured and uploaded separately.